COHOMOLOGY OF AN OBJECT OF A RINGED SITE EQUALS COHOMOLOGY ON THE LOCALIZED SITE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the representative, the chain of equalities and the K-injectivity of the restriction are all in the counted span.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space. That statement is the special case of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Cohomology H^p(U, K) of an object of the derived category at an object of the site. Same chapter, the definition of H^q(U, K) as the cohomology of the complex of sections Gamma(U, I^bullet) for a K-injective representative. Supplies both sides of the asserted equality, since cohomology is computed by these section complexes.
- Localized site C/U and restriction of a sheaf to it. Same chapter, the construction of the localized site C/U and the restriction K|_{C/U} of an object of D(O). Supplies the site on the right hand side and the identification of sections over C/U with sections over U.
- K-injectivity of the restriction of a K-injective complex. Same chapter, the result that restricting a K-injective complex of O-modules to a localized site keeps it K-injective. Supplies the final step, which makes the section complex over C/U compute the cohomology of the restriction.

Source reference. Original lemma 08FI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4314–4314. Same-chapter navigation pointer invoked at proof line 4343, where the author uses that the restriction of a K-injective complex to the localized site is again K-injective. The pointer only resolves that reference for the reader; the cited lemma is not reproduced inside the counted span, and no bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $U \in \mathrm{Ob}(\mathcal{C})$. For K in $D(\mathcal{O})$ we have $H^p(U, K) = H^p(\mathcal{C}/U, K|_{\mathcal{C}/U})$.*

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Proof. Let $\mathcal{I}^\bullet$ be a K-injective complex of $\mathcal{O}$-modules representing K. Then

$$H^q(U, K) = H^q(\Gamma(U, \mathcal{I}^\bullet)) = H^q(\Gamma(\mathcal{C}/U, \mathcal{I}^\bullet|_{\mathcal{C}/U}))$$

by construction of cohomology. By Lemma 08FI the complex $\mathcal{I}^\bullet|_{\mathcal{C}/U}$ is a K-injective complex representing $K|_{\mathcal{C}/U}$ and the lemma follows. $\square$